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Language Models (Mostly) Know What They Know

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Abstract

We study whether language models can evaluate the validity of their own claims and predict which questions they will be able to answer correctly. We first show that larger models are well-calibrated on diverse multiple choice and true/false questions when they are provided in the right format. Thus we can approach self-evaluation on open-ended sampling tasks by asking models to first propose answers, and then to evaluate the probability "P(True)" that their answers are correct. We find encouraging performance, calibration, and scaling for P(True) on a diverse array of tasks. Performance at self-evaluation further improves when we allow models to consider many of their own samples before predicting the validity of one specific possibility. Next, we investigate whether models can be trained to predict "P(IK)", the probability that "I know" the answer to a question, without reference to any particular proposed answer. Models perform well at predicting P(IK) and partially generalize across tasks, though they struggle with calibration of P(IK) on new tasks. The predicted P(IK) probabilities also increase appropriately in the presence of relevant source materials in the context, and in the presence of hints towards the solution of mathematical word problems. We hope these observations lay the groundwork for training more honest models, and for investigating how honesty generalizes to cases where models are trained on objectives other than the imitation of human writing.

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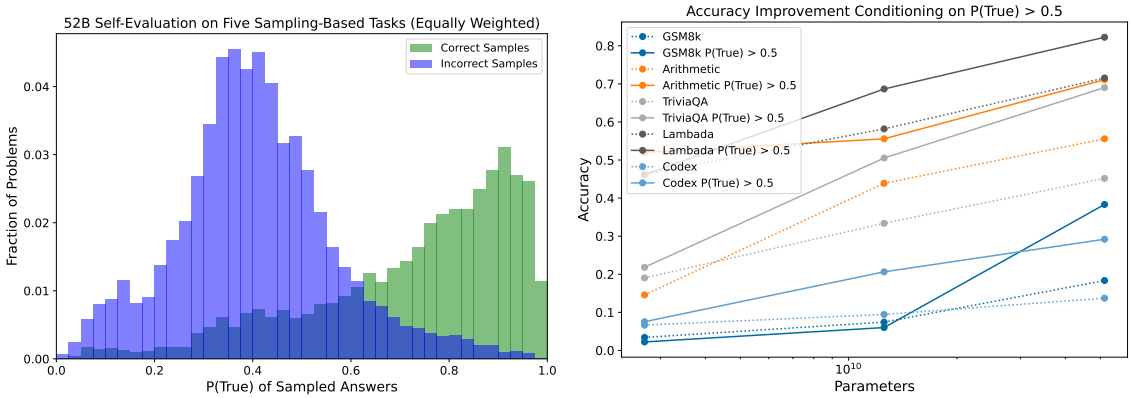


Figure 1 **(left)** We show the overall ability of a 52B language model to evaluate its own proposed answers (sampled at unit temperature) to questions from TriviaQA, Lambada, Arithmetic, GSM8k, and Codex HumanEval. We have weighted the overall contribution from each of these five datasets equally. We evaluate 20-shot using the method of section 4, where we show the model several of its own samples and then ask for the probability P(True) that a specific sample is correct. **(right)** We show the improvement in the accuracy on each sampling task when only including questions where a randomly sampled (unit temperature) response achieved P(True) > 0.5.

1 Introduction

We would eventually like to train AI systems that are honest, which requires that these systems accurately and faithfully evaluate their level of confidence in their own knowledge and reasoning. So AI systems must be able to recognize what they do and do not know, as a prerequisite. In this work, we study the extent to which Language Models (LMs) possess this ability and how it can be elicited and imparted.

As a starting point, we examine calibration: do the probabilistic predictions from language models match up with frequencies of occurrence? Language models can produce well-calibrated predictions for token probabilities on-distribution [Guo et al., 2017]. We show that large language models are also well-calibrated on a diverse array of multiple choice questions, as long as the questions are formatted appropriately. In particular, calibration improves with model size and few-shot prompting.

Good calibration opens up the possibility for using models to evaluate the accuracy of their own outputs ("self-evaluation"). For example, given any open-ended query, we can sample an answer from the model and then have the model evaluate P(True), the probability that its answer is correct. We may expect self-evaluation to be challenging, because the model may be overconfident that its own samples[1] are correct. Our self-evaluation procedure nevertheless distinguishes correct and incorrect samples, as summarized in Figure 1. Furthermore, as model size and capabilities increase, models improve at self-evaluation, which suggests that verification improves faster than generation quality in this context.

We also show that self-evaluation can be improved if we provide a model with *many* of its own samples, before asking it to evaluate any single sample. That is, 'brainstorming other possibilities' helps large models to evaluate the validity of a given answer option.

These techniques address a question about the world, as they ask models to evaluate "according to accepted truth in the wider world (i.e. according to humans), is a particular answer to a question correct?" In the case of self-evaluation, the proposed answer was provided by the model, but its validity is nevertheless an external fact.

But we are also interested in having language models attempt to directly evaluate their own state of knowledge. To this purpose, we investigate what happens when we train models to predict whether or not they can correctly answer questions themselves. This is really a question about the model[2] since we are training the model to learn what sorts of questions *it*, in particular, can answer.

[1] Another more subtle issue is that in our setup, models have no way to distinguish the tokens they've written from tokens they were given by a third party.

[2] This claim is subtle, since the model could be learning "how hard is this question in general?" rather than "do I know the answer to this?" We partially address this concern in section 5.5.